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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr.Super60_NUCLEUS & STERLING_BT Paper -2(Adv-2022-P2-Model)

Date: 21-07-2024

Time: 02.00Pm to 05.00Pm

CTA-01

Max. Marks: 180

21-07-2024_Sr.Super60_NUCLEUS & STERLING_BT_Jee-Adv (2022-P2)_CTA-01_Syllabus

PHYSICS

: Thermal physics, Heat Transfer, Ray Optics: Rectilinear propagation of light; Reflection, Magnification.

CHEMISTRY

: Nomenclature, Isomerism, GOC, Alkanes

MATHEMATICS

: Function and Inverse Trigonometric Function, LCD

Name of the Student: _____

H.T. NO:

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JEE-ADVANCED-2022-P2-Model

Time:3Hr's

IMPORTANT INSTRUCTIONS

Max Marks: 180

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 8)	Questions with Integer Answer Type	+3	-1	8	24
Sec – II(Q.N : 9 – 14)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 15 – 18)	Questions with Single Correct Choice	+3	-1	4	12
Total				18	60

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 26)	Questions with Integer Answer Type	+3	-1	8	24
Sec – II(Q.N : 27 – 32)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 33 – 36)	Questions with Single Correct Choice	+3	-1	4	12
Total				18	60

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 44)	Questions with Integer Answer Type	+3	-1	8	24
Sec – II(Q.N : 45 – 50)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 51 – 54)	Questions with Single Correct Choice	+3	-1	4	12
Total				18	60

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THE PERFECT HAT-TRICK WITH ALL-INDIA RANK 1
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023

JEE MAIN 2023
 SINGARAJU VENKAT KOUNDINYA
 Rank 1
 300/300
 AIR 1



RANK 1

JEE Advanced 2023
 VAVILALA CHIDVILAS REDDY
 Rank 1
 341/360
 AIR 1



RANK 1

NEET 2023
 BORA VARUN CHAKRAVARTHI
 Rank 1
 720/720
 AIR 1



RANK 1



MATHEMATICS

Max Marks: 60

SECTION-I
(INTEGER ANSWER TYPE)

- This section contains EIGHT (08) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If ONLY the correct integer is entered;

Zero Marks: 0 If the question is unanswered;

Negative Marks: -1 In all other cases

1. Find the number of elements contained in the range of the function

$$f(x) = \left\lfloor \frac{x}{6} \right\rfloor \left\lceil \frac{-6}{x} \right\rceil \forall x \in (0, 30] \text{ (where } \lfloor \cdot \rfloor \text{ denotes greatest integer function)}$$

2. If $f: \mathbb{R} \rightarrow (-\infty, -1]$ be a function defined by

$$f(x) = (ab + 2a - b - 2)x^5 - (a^2 - 2a + 1)x^3 + (a^2 - 2a - 3)x^2 + (a^2 + 2b)x - 5, a, b \in \mathbb{R} \text{ is surjective, then the absolute value of sum of the all possible values of } b \text{ is } \underline{\hspace{2cm}}$$

$$\left(e \left(\frac{1}{1+x} \right)^{\frac{1}{x}} - 1 \right)$$

3. If $\lim_{x \rightarrow 0^+} \frac{\left(e \left(\frac{1}{1+x} \right)^{\frac{1}{x}} - 1 \right)}{x} = \ell$, then 4ℓ is equal to

4. Find the number of values of $\alpha \in [0, 1]$ satisfying

$$4\alpha(1-\alpha) = \beta, \quad \gamma = 4\beta(1-\beta), \quad \frac{2024\alpha}{2025} = 4\gamma(1-\gamma).$$

5. If $\min_{p, q, r, s \in \mathbb{R}^+} \left(\sqrt{p^2 + (20-q)^2} + \sqrt{q^2 + (21-r)^2} + \sqrt{r^2 + (20-s)^2} + \sqrt{s^2 + (21-p)^2} \right) = \lambda$, then the value of $\lambda - 53$ is equal to

6. If $f(x) = \prod_{k=1}^{\infty} \left(\frac{1 + 2 \cos \left(\frac{2x}{3^k} \right)}{3} \right)$, then number of points where

$$\left[x f(x) \right] + |x f(x)| + (x-1) \left| x^2 - 3x + 2 \right| \text{ is non-differentiable in } x \in (0, 14\pi) \text{ is equal to 'n' where (n - 18) is equal to (where } \lfloor \cdot \rfloor \text{ denotes greatest integer function)}$$

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JEE MAIN
2023

SINGARAJU
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MAINS

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350
MAINS

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JEE Prep Class
720
720
MAINS

RANK
1



7. Let $f: R \rightarrow R$, $f(x) = x^3 + 3x + 1$. If inverse of the function can be expressed as
- $$f^{-1}(x) = \left(\frac{x - a + \sqrt{x^2 - bx + c}}{2} \right)^{1/3} + \left(\frac{x - a - \sqrt{x^2 - bx + c}}{2} \right)^{1/3}$$

Then the value of

$$2a + 3b = \dots\dots\dots$$

8. If $f(x) = \lim_{n \rightarrow \infty} \tan^{-1} \left(4n^2 \left(1 - \cos \frac{x}{n} \right) \right)$ and $g(x) = \lim_{n \rightarrow \infty} \frac{n^2}{2} \ln \cos \left(\frac{2x}{n} \right)$, then

$$\lim_{x \rightarrow 0} \frac{e^{-2g(x)} - e^{f(x)}}{x^6} = k, \text{ where } 3k = \dots\dots\dots$$

SECTION – II (ONE OR MORE CORRECT ANSWER TYPE)

•This section contains SIX (06) questions.

•Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

•For each question, choose the option(s) corresponding to (all) the correct answer(s).

•Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks: +4 If only (all) the correct option(s) is(are) chosen; **Partial Marks:** +3 If all the four options are correct but ONLY three options are chosen;

Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;

Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;

Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks: -2 In all other cases.

9. Which of the following is/are correct?

A) Let a function f defined from $R \rightarrow R$ as $f(x) = \begin{cases} m - x, & \text{for } x \leq 1 \\ 2mx + 1, & \text{for } x > 1 \end{cases}$. If the function f is

onto then the range of m is $[-2, 0)$

B) Let $g: R \rightarrow R$, such that $g(x) = \ln(\sqrt{1+x^6} - x^3)$ then g is one – one and onto

C) Let, $f: R \rightarrow R$ such that $f(x) = \left[x + \frac{1}{2} \right] + \left[x - \frac{1}{2} \right] + 2[-x]$ (where $[.]$ denotes G.I.F)

then $f(x)$ is many one and into

D) Let, $f: R \rightarrow R$, such that $f(x) = e^{\sin\{x\}} + \sin\left(\frac{\pi}{2}[x]\right)$ (where $[.]$ denotes G.I.F, $\{.\}$

denote fractional part function) then $f(x)$ is many one and into

10. If $f: [a, b] \rightarrow [5, 6]$ be defined as $y = f(x) = 2x^3 - 9x^2 + 12x + 1$ then which of the following is/are TRUE?

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**JEE MAIN
2023**

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300
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RANK

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**JEE Advanced
2023**

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Super 60 Class
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RANK

1

**NEET
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Rank 1
Sri Chaitanya
Super 60 Class
720
720
MAHAR



RANK

1



- A) Minimum value of a for $y = f(x)$ to be a one-one function is $\frac{1}{2}$
- B) Minimum value of a for $y = f(x)$ to be a one-one function is 2
- C) Minimum value of b for $y = f(x)$ to be an onto function is 1
- D) Maximum value of b for $y = f(x)$ to be an onto function is $\frac{5}{2}$

11. If $f(x) = \frac{[x]+1}{\{x\}+1}$ for $f: \left[0, \frac{5}{2}\right) \rightarrow \left(\frac{1}{2}, 3\right]$. Then which of the following is/ are true?

(where $[.]$ denotes greatest integer function and $\{.\}$ denotes fractional part function)

- A) $f(x)$ is injective, discontinuous function
- B) $f(x)$ is surjective, non differentiable function
- C) $\min\left(\lim_{x \rightarrow 1^-} f(x), \lim_{x \rightarrow 1^+} f(x)\right) = f(1)$
- D) Minimum value of $f(x) = \frac{1}{2}$

12. Let $f(x) = \begin{cases} \frac{\frac{1}{e^x} + \frac{2}{e^x} + \frac{3}{e^x}}{p.e^{\left(-2+\frac{1}{x}\right)} + q.e^{\left(-1+\frac{3}{x}\right)}} & 0 < x < \frac{\pi}{2} \\ (\sin x + \cos x)^{\cos ex} & \frac{-\pi}{2} < x < 0 \end{cases}$ and $\lim_{x \rightarrow 0} f(x)$ exists and finite then

correct options can be:

- A) $p = 2, q = 1$
- B) $p + q$ can be equal to 6
- C) $p \in R, q = 1$
- D) $p = 3, q = 1$

13. Let $f(x) = \begin{cases} \sin \pi x; & 0 \leq x \leq 2 \\ 2 - x; & 2 < x \leq 5 \end{cases}, g(x) = \begin{cases} \max(|x|, 1 - |x|, |x - 1|); & \text{If } x \in (0, 1) \\ \min(|x|, 1 - |x|, |x - 1|); & \text{otherwise} \end{cases}$

then which of the following is/are TRUE?





A) $f\left(g\left(\frac{1}{4}\right)\right) = \frac{1}{\sqrt{2}}$

B) $f(g(-1)) = 1$

C) $f(|g(-2)|) = 0$

D) $f\left(g\left(\frac{3}{4}\right)\right) = \frac{1}{\sqrt{2}}$

14. Let $f: A \rightarrow B$ be an onto function defined as $f(x) = \frac{\sin^{-1}x + \tan^{-1}x}{\cos^{-1}x + \cot^{-1}x}$. Then which of

the following is/are **TRUE**?

A) If the minimum and maximum value of $f(x)$ be m and M respectively then the value of

$\left|\frac{M}{m}\right|$ is 7

B) The number of solutions of the equations $f(x^3 + 14x^2 + 13x - 5) = f(1 - x^2 + x^3)$ is 1

C) If the minimum and maximum value of $f(x)$ be m and M respectively then the value of

$\left|\frac{M}{m}\right|$ is 8

D) The number of solutions of the equations $f(x^3 + 14x^2 + 13x - 5) = f(1 - x^2 + x^3)$ is 2

SECTION – III (SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -1 In all other cases

15. If $\sin^{-1}x, \cos^{-1}x$ are inverse circular function. The principal values of above function are given below in their respective domain: $\sin^{-1}x \in \left[\frac{3\pi}{2}, \frac{5\pi}{2}\right], \cos^{-1}x \in [2\pi, 3\pi]$. If

$\sin^{-1}\sin(10) = a\pi + b$ (where $a, b \in I$) then $|a| + |b|$ is equal to :

A) 13

B) 14

C) 12

D) 15

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**THE PERFECT HAT-TRICK WITH ALL-INDIA RANK 1
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**JEE MAIN
2023**

SINGARAJU
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Rank 1st
Sri Chaitanya
Super 60 Class
300
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MAINS



**RANK
1**

**JEE Advanced
2023**

VARILALA
CHIDVILAS REDDY
Rank 1st
Sri Chaitanya
Super 60 Class
341
350
MAINS



**RANK
1**

**NEET
2023**

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CHAKRAVARTHI
Rank 1st
Sri Chaitanya
Super 60 Class
720
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MAINS



**RANK
1**



16. Determine the number of distinct integers among the numbers

$$\left[\frac{1^2}{2024} \right], \left[\frac{2^2}{2024} \right], \dots, \left[\frac{2023^2}{2024} \right] \quad ([.] \text{ denotes G.I.F})$$

- A) 1516 B) 1517 C) 1518 D) 1519

17. Let $f(x) = 8x^3 + 3x$ and $f^{-1}(x)$ be the inverse function of $f(x)$, then

$$\lim_{x \rightarrow \infty} \frac{\frac{1}{x^3}}{f^{-1}(8x) - f^{-1}(x)} =$$

- A) 0 B) 1 C) 2 D) 4

18. If $f(x)$ is a real polynomial function and $(f(\alpha))^2 + (f'(\alpha))^2 = 0$ ($\alpha \in R$) then find

$$\lim_{x \rightarrow \alpha} \frac{f(x)}{f'(x)} \left[\frac{f'(x)}{f(x)} \right] \quad (\text{when } [.] \text{ greatest integer function})$$

- A) 0 B) 1 C) -1 D) Can't be determined

JEE MAIN
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JEE Prep Class
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2023

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CHIDVILAS REDDY
AIR 1
Sri Chaitanya
JEE Prep Class
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RANK

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NEET
2023

BORA VARUN
CHAKRAVARTHI
AIR 1
Sri Chaitanya
JEE Prep Class
720
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RANK

1



PHYSICS

Max Marks: 60

SECTION-I (INTEGER ANSWER TYPE)

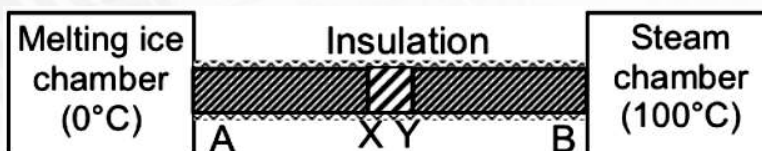
- This section contains EIGHT (08) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks: +3 If ONLY the correct integer is entered;

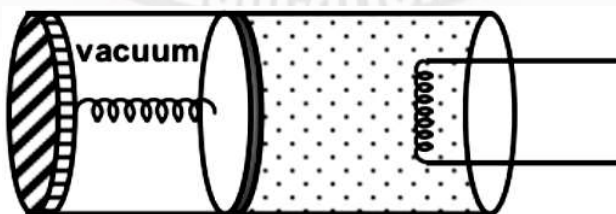
Zero Marks: 0 If the question is unanswered;

Negative Marks: -1 In all other cases

19. A conducting rod is initially at 0°C temperature. Now one end A is kept at melting ice chamber (0°C) and the other end B is kept at steam chamber (100°C). The surface of the rod is well insulated from the outer environment so that heat is only flowing along the length of the rod as shown in the figure. The cross-sectional area of the rod is 1 cm^2 and length of the rod is 1 m. Thermal conductivity of the rod is 500 W/m-K . Density of the rod is 8000 kg/m^3 . Specific heat capacity of the rod is 500 J/kg-K . A small cylindrical element xy (of length 1 cm and cross-sectional area same as that of rod) is examined. It is observed that temperature gradient of the end X is 5°C/m and average temperature of xy part of the rod is rising at 0.05°C/sec . What is the temperature gradient (in $^\circ\text{C/m}$) of end Y of the element?



20. Monoatomic gas is kept in equilibrium in an insulated cylinder of length L. Piston is attached with an ideal spring as shown which has relaxed length L. The spring is fixed at one end of cylinder. The gas is supplied very small amount of heat dQ . Piston gets very small displacement in this process. The work done by gas is $\frac{dQ}{N}$. The value of N is



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**THE PERFECT HAT-TRICK WITH ALL-INDIA RANK 1
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JEE MAIN
2023SINGARAJU
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AIR 1
Sri Chaitanya
JEE Prep Class300
300
MAINS

RANK

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2023VANILALA
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AIR 1
Sri Chaitanya
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RANK

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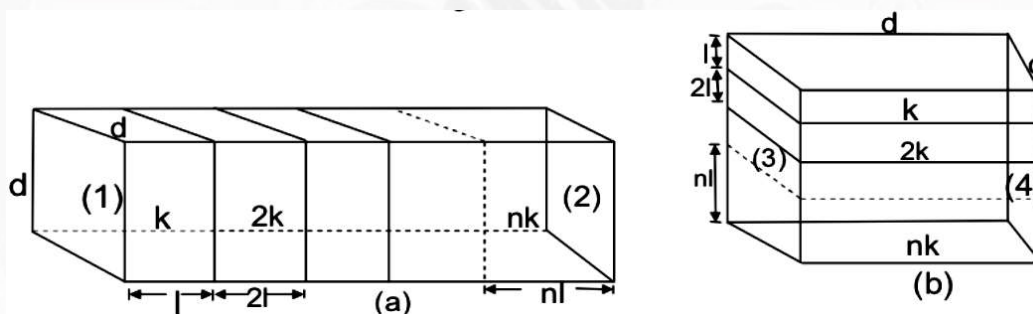
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2023BORA VARUN
CHAKRAVARTHI
AIR 1
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JEE Prep Class720
720
MAINS

RANK

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21. Two rods each of length L_1 and one rod of length L_2 for isosceles triangle having base of length L_2 . Co-efficient of linear expansion of base is $4 \times 10^{-6} / ^\circ C$ while for rod of length L_1 is $25 \times 10^{-6} / ^\circ C$. Find the ratio $\left(\frac{L_2}{L_1}\right)$, if the length of latitude remains same when the temperature of the system is increased by $10^0 C$.
22. An arrangement consisting of slabs is shown in the figure. When system is placed in situation (a) with temperature difference is maintained between (1) and (2), equivalent thermal conductivity of system is K and when system is changed to situation (b) and temperature difference is maintained between (3) and (4) then equivalent thermal conductivity increases by 20%. Find number of slabs in the arrangement.



23. In a thermally isolated cylinder, a piston can move without friction up and down. The cylinder is divided into two compartments by a freely moving weightless wall which conducts slowly heat, see figure. Initially the gas temperatures are equal, and the volume of hydrogen is 3 times smaller than that of helium. Helium receives a certain amount of heat, because of which the piston moves up by $d_1 = 11 \text{ cm}$. After waiting for some longer period of time, an additional displacement of the piston was observed. By how much and in what direction the piston moved additionally? Gases can be considered to be ideal.

Molar heat capacitance at constant pressure is $C_{PH_2} = \frac{7R}{2}$ for hydrogen, and

$$C_{PHe} = \frac{5R}{2} \text{ for helium.}$$



JEE MAIN 2023

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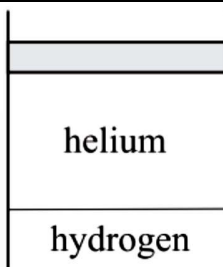
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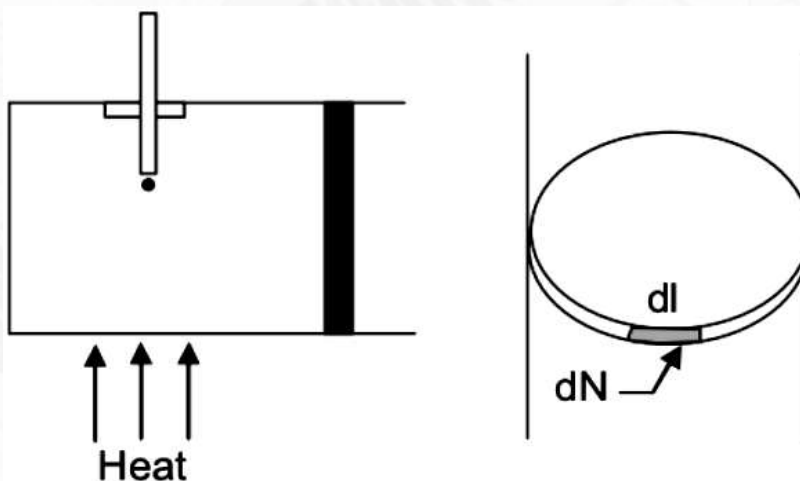
BORA VARUN
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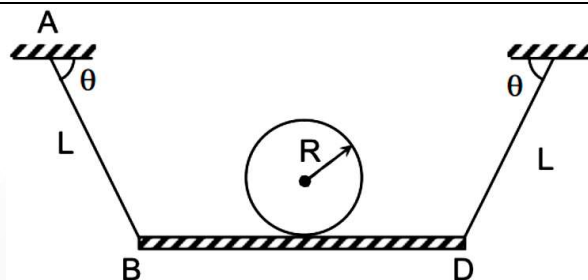


24. Figure shows a cylindrical tube of radius 5 cm and length 20 cm. It is closed by tight fitting cork. The friction coefficient between the cork and the tube is 0.20. The tube contains an ideal gas at a pressure of 1 atm and temperature 300K. The tube is slowly heated and it is found that the cork pops out when the temperature reaches 600K. Let dN denote the magnitude of normal contact force exerted by small length $d\ell$ of the cork along the periphery. Assuming that the temperature of gas is uniform at any instant, the value of $\frac{dN}{d\ell} = 25 \times \alpha \times 10^\beta$. Find the value of $\alpha + \beta$.

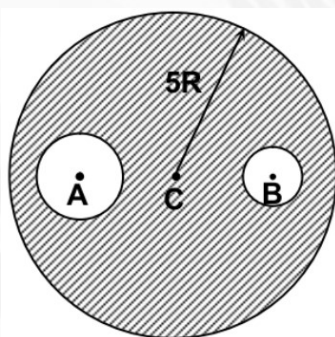


25. A metal cylinder of radius R is placed on a wooden plank BD . The plank is kept horizontal suspended with the help of two identical strings AB and CD each of length L . The temperature coefficient of linear expansion of the cylinder and the strings are α_1 and α_2 respectively. Angle θ shown in the figure is 30° . It was found that with change in temperature the centre of the cylinder did not move. Find the ratio $\frac{\alpha_1}{\alpha_2}$, if it is known that $L = 4R$. Assume that change in value of θ is negligible for small temperature changes.





26. A liquid drop of radius $5R$ is kept in the open atmosphere of pressure P_0 . C is the centre of the liquid drop. Two air bubbles are formed inside the liquid drop centred at A and B of radii $2R$ and R respectively. Let the gauge pressure at A , B and C are P_A , P_B and P_C respectively. The ratio $\frac{P_A P_B}{P_C}$ is equal to 3λ . Find the value of λ .



SECTION – II (ONE OR MORE CORRECT ANSWER TYPE)

•This section contains SIX (06) questions.

•Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

•For each question, choose the option(s) corresponding to (all) the correct answer(s).

•Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks: +4 If only (all) the correct option(s) is(are) chosen; *Partial Marks:* +3 If all the four options are correct but ONLY three options are chosen;

Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;

Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;

Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks: -2 In all other cases.

27. In a thermodynamic process Helium gas obeys the law $TP^{-2/5}$ constant. If temperature of 2 moles of the gas is raised from T to $3T$, then
- A) Heat given to the gas is $9RT$
- B) Heat given to the gas zero
- C) Work done by the gas is $-6RT$
- D) $\int VdP$ for the process in the given temperature range is $10RT$

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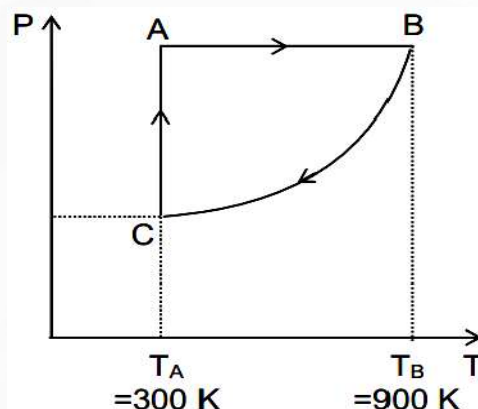


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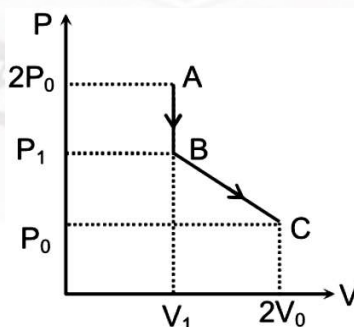
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28. Two moles of an ideal monoatomic gas is taken through a cyclic process ABCA as shown in the P-T diagram. In the process BC, $P = \alpha T^2$, where α is a constant. Then choose the correct option(s). (R = ideal gas constant) [Take $\ln(3) = 1.1$]



- A) The heat absorbed by the gas during the process AB is $3000R$.
 B) The heat absorbed by the gas during the process BC is $1200R$.
 C) The heat rejected by the gas during the process BC is $600R$.
 D) The efficiency of the cycle is 36%.
29. During the melting of a slab of ice at 273K at the atmospheric pressure.
- A) Positive work is done by the ice-water system on the atmosphere.
 B) Positive work is done on the ice-water system by the atmosphere.
 C) The internal energy of the ice-water system increases.
 D) The internal energy of the ice-water system decreases.
30. A monoatomic gas consisting of n moles is taken from state A to state C as shown in the figure. Process $A \rightarrow B$ is isochoric which of the following is/are **CORRECT**?


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 Rank 1
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 AIR 1

RANK
1



A) Temperature of gas at state B is minimum for $P_1 = \frac{3P_0}{2}$ and $V_1 = \frac{3V_0}{2}$.

B) For $P_1 = \frac{3P_0}{2}, V_1 = \frac{3V_0}{2}$, minimum internal energy of gas will be $3P_0V_0$.

C) For $P_1 = \frac{7P_0}{4}, V_1 = \frac{3V_0}{2}$, heat absorbed/released from $A \rightarrow C$ will be more compare to heat released/absorbed from $A \rightarrow C$ when $P_1 = \frac{3P_0}{2}, V_1 = \frac{7V_0}{4}$.

D) In process $B \rightarrow C$, temperature of gas first decreases and then increases if $P_1 = \frac{3P_0}{2}, V_1 = \frac{3V_0}{2}$.

31. 10g of ice at 0°C is mixed with 5g of steam at 100°C . If latent heat of fusion of ice is $80\text{cal} - \text{g}^{-1}$ and latent heat of vaporization of $540\text{cal} - \text{g}^{-1}$. When thermal equilibrium is reached then which of the following is/are **CORRECT**?

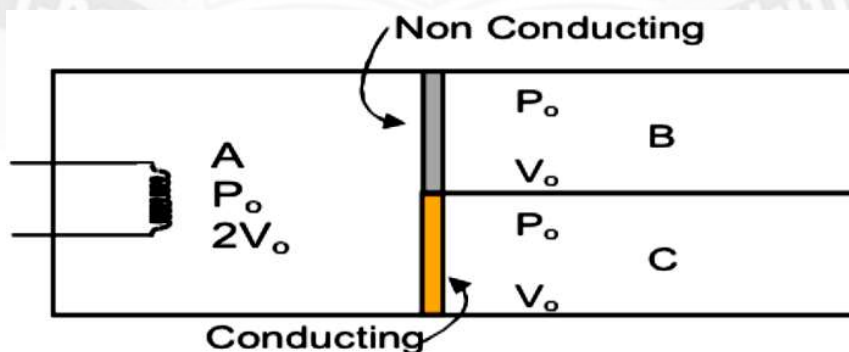
A) Temperature of the mixture is 0°C

B) Temperature of mixture is 100°C

C) Mixture contains 13.3 g of water and 1.67 g of steam

D) Mixture contains 5.3 g of ice and 9.7 g of water

32. An adiabatic container of volume $4V_0$ is divided into two equal parts by rigid fixed wall whose lower half is conducting and upper half in non-conducting. Initially the right portion is divided into two equal parts by a freely moving massless and non conducting piston. Section A has 2 moles, B and C each has one mole of an ideal gas with adiabatic exponent $(\gamma)1.5$. If the heater in left part is switched on till the final pressure C becomes $(125/27)P_0$. Mark the **CORRECT** statements.





- A) Final temperature in section A $\frac{205P_0V_0}{27R}$
- B) Final temperature in section B $\frac{5P_0V_0}{3R}$
- C) Final temperature in section C $\frac{205P_0V_0}{27R}$
- D) Heat supplied by the heater $\frac{368}{9}P_0V_0$

SECTION – III (SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

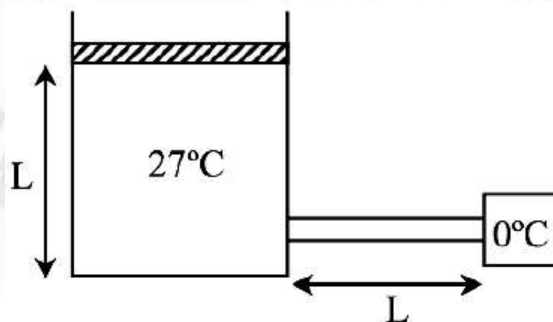
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -1 In all other cases

33. 0.5 mole of an ideal gas at constant temperature 27^0C kept inside a cylinder of length L and cross section area A closed by a mass less piston. The cylinder is attached with conducting rod length L cross section area $\left(\frac{1}{9}\right)m^2$ and thermal conductivity K , whose other end is maintained at 0^0C . If piston is moved such that rate of heat flow through the conducting rod is constant then velocity of piston when it is at height $\frac{L}{2}$ from the bottom of cylinder [neglect any kind of heat loss from system]



- A) $\frac{K}{R}m/s$ B) $\frac{K}{10R}m/s$ C) $\frac{K}{100R}m/s$ D) $\frac{K}{1000R}m/s$

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Space for rough work

Page 14



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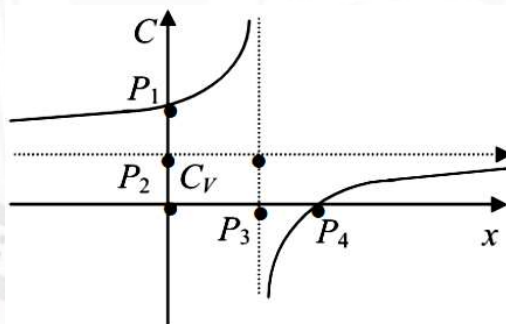
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34. A spherical black body has a radius R and steady surface temperature T , heat sources ensure the heat evolution at a constant rate and distributed uniformly over its volume. What should be the new steady state surface temperature of the object if the radius is decreased by half? Assume surrounding to be at absolute zero and heat evolution rate through unit volume remain same.
- A) $(2)^{\frac{1}{4}} T$ B) $\sqrt{2} T$ C) $\frac{T}{(2)^{\frac{1}{4}}}$ D) $\frac{T}{\sqrt{2}}$
35. The relation between internal energy U , pressure P and volume V of a gas in an adiabatic process $U = a + b PV$, where a and b are positive constants. Then
- A) Equation for the process is $PV^{\frac{b+2}{b}} = \text{constant}$
- B) Equation for the process is $PV^{\frac{b+1}{b}} = \text{constant}$
- C) The gas must be monoatomic
- D) The gas must be diatomic
36. The following graph gives the variation of molar heat capacity C a function of x where x is given by the expression $PV^x = \text{constant}$. If gas is monoatomic what are the coordinates of P_1 and P_4 .



- A) $\left(0, \frac{5}{2}R\right) \& \left(\frac{5}{3}, 0\right)$ B) $\left(0, \frac{3}{2}R\right) \& \left(\frac{7}{5}, 0\right)$
- C) $\left(0, \frac{5}{2}R\right) \& \left(\frac{7}{5}, 0\right)$ D) $\left(0, \frac{7}{5}R\right) \& \left(\frac{5}{3}, 0\right)$





CHEMISTRY

Max Marks: 60

SECTION-I
(INTEGER ANSWER TYPE)

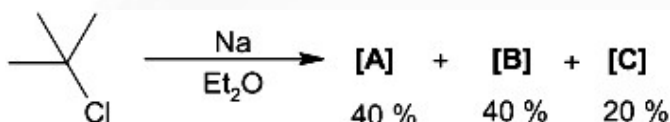
- This section contains EIGHT (08) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If ONLY the correct integer is entered;

Zero Marks: 0 If the question is unanswered;

Negative Marks: -1 In all other cases

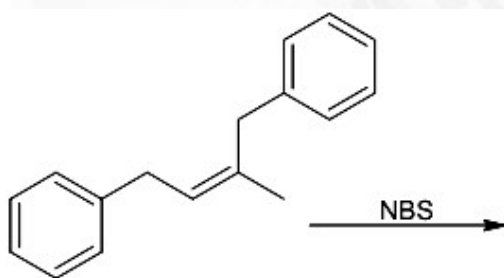
37. Consider the following reaction:



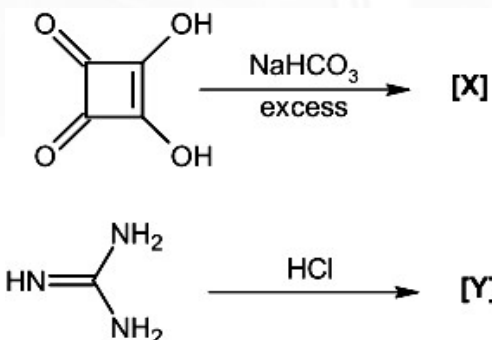
[A], [B] & [C] are all organic products.

The number of sp^3 carbon atoms in [C] is _____.

38. The number of monobromoproducts (excluding stereoisomers) formed in the following reaction is _____.



39. Consider the organic products [X] and [Y] of the following reactions:



If the number of equivalent contributing structures of [X] is x and [Y] is y , the value of $(x - y)$ is _____.



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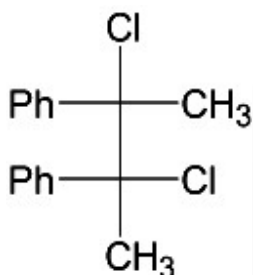


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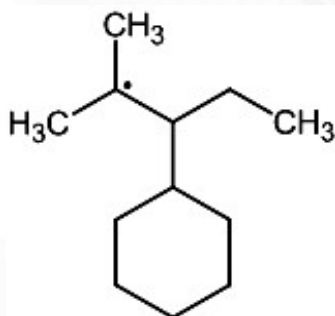
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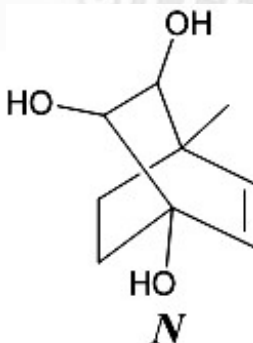
40. The total number of stable conformers with zero dipole moment for the following compound is



41. The magnitudes of enthalpy of combustions of CH_4 , C_2H_6 , C_3H_8 and n -butane in $kJ\ mol^{-1}$ are 890, 1560, 2220 & 2879 respectively. The ratio of enthalpies of combustion of n -heptane to that of ethane rounded off to nearest integer is _____.
42. The number of isomers of the formula C_3H_6O including stereoisomers but excluding tautomers is _____.
43. The total number of contributing structures showing hyperconjugation (involving $C-H$ bonds) for the following free radical is



44. The total number of optically active isomers that exist for N is _____.





SECTION – II

(ONE OR MORE CORRECT ANSWER TYPE)

• This section contains SIX (06) questions.

• Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

• For each question, choose the option(s) corresponding to (all) the correct answer(s).

• Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks: +4 If only (all) the correct option(s) is(are) chosen; **Partial Marks:** +3 If all the four options are correct but **ONLY** three options are chosen;

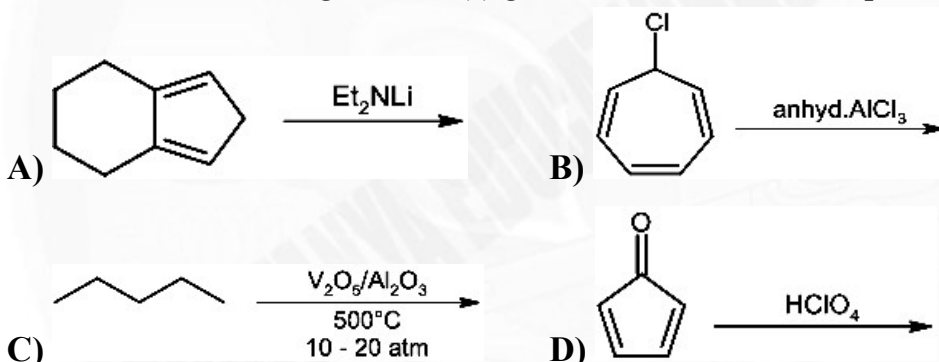
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;

Partial Marks: +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;

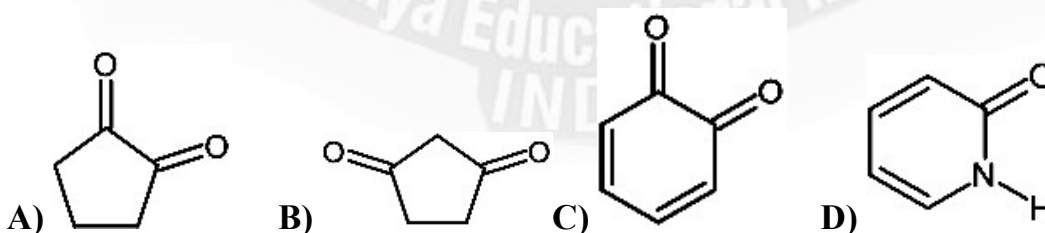
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks: -2 In all other cases.

45. Which of the following observation(s) can be accounted for using hydrogen bonding?
- A) Thiophenol is more acidic than phenol.
 B) Carboxylic acids exist as dimers in vapour state.
 C) 2-nitrophenol is less acidic than 4-nitrophenol.
 D) Pentan-1-ol has higher boiling point than butan-1-ol
46. Which of the following reaction(s) generate an aromatic compound as the major product?



47. The correct combination of names for isomeric amines with molecular formula $\text{C}_4\text{H}_{11}\text{N}$ is/are
- A) tert-butanamine and 2-methylpropan-2-amine
 B) n-butanamine and butan-1-amine
 C) tert-butanamine and 1,1-dimethylethan-1-amine
 D) iso-butyl amine and 2-methylpropan-1-amine
48. Tautomerism is exhibited by



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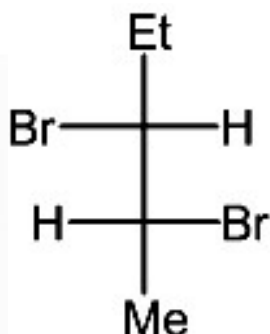
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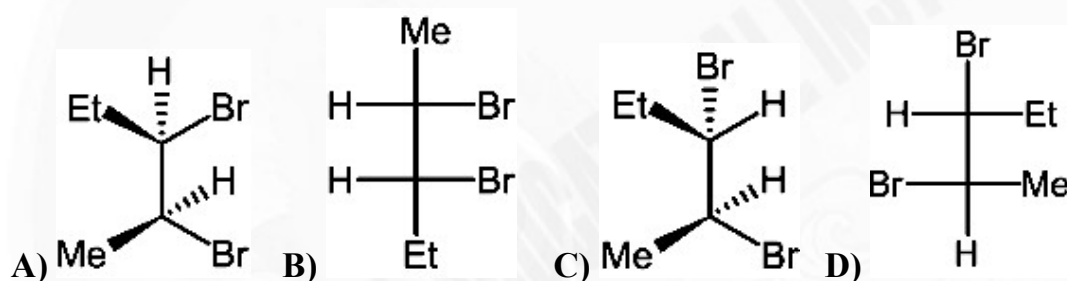
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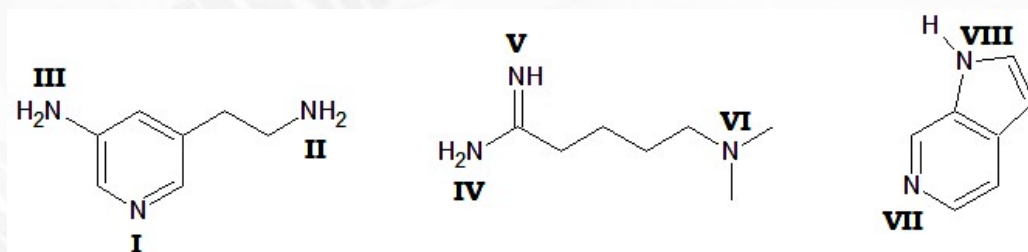
49. Given below is one of the *threo*-isomer of 2,3-dibromopentane:



Choose the structure(s) of its diastomer(s):



50. Consider the following compounds and the nitrogen atoms labelled from I to VIII.



Choose the correct statement(s) from the following:

- A) If an equimolar mixture of all these three compounds is treated with HCl, V gets protonated first.
- B) Basic character of I, II and III follow the order II>I>III
- C) The lower basic nature of VIII compared to VII can be attributed to involvement of lone pair on VIII in aromatic stabilization.
- D) Basic character of I, V and VII will almost be equal as all them are sp^2 hybridized.





SECTION – III (SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

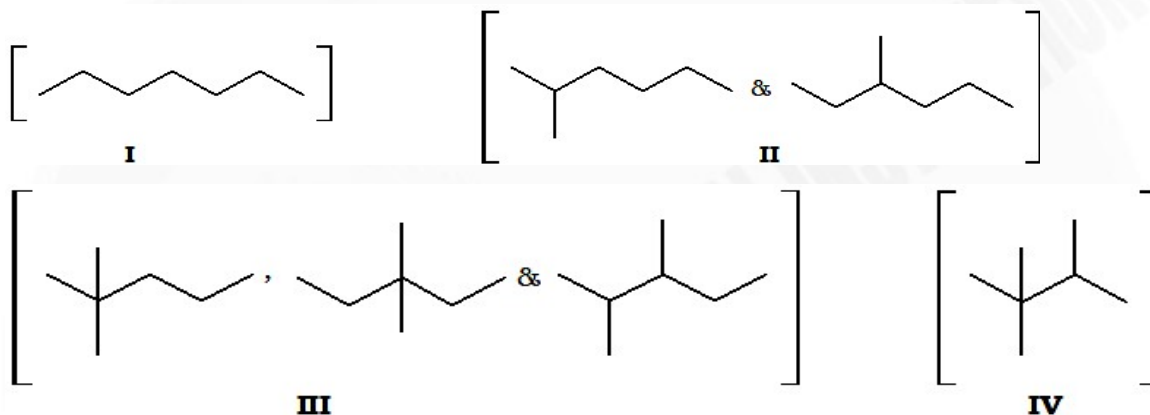
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -1 In all other cases

51. Few isomers of heptane are classified based on branching into the following categories:



Choose the correct order(s) of physical properties of these sets of compounds:

P) Boiling Point: I>II>III>IV

Q) Melting Point: I<II<III<IV

R) Heat of combustion: IV<III<II<I

S) Stability: IV>III>II>I

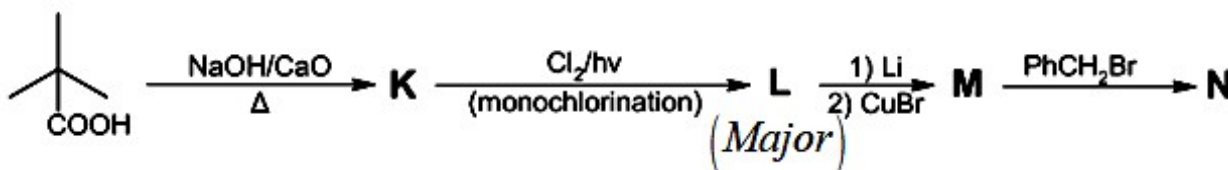
A) PQ only

B) QRS only

C) QS only

D) PQRS

52. Consider the following reaction sequence:



Choose the correct option

A) N has 9 primary hydrogen atoms.

B) K reacts with KMnO_4 to give isobutyric acid.

C) N on reaction with Br_2/light gives 7 isomeric products of which only one is achiral.

D) All hydrogen atoms in M are primary.



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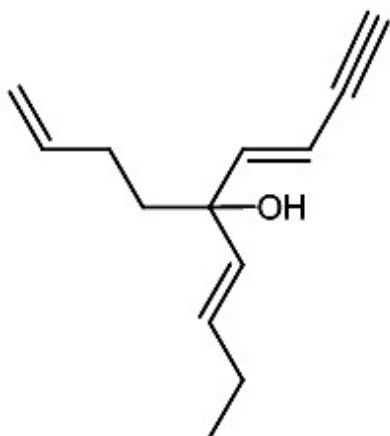
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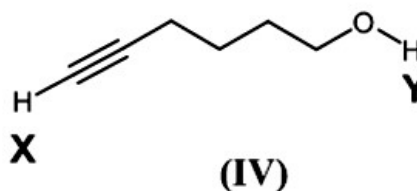
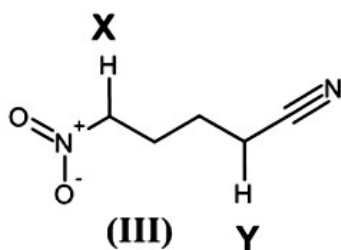
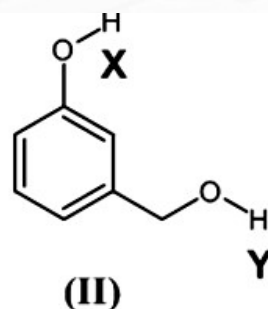
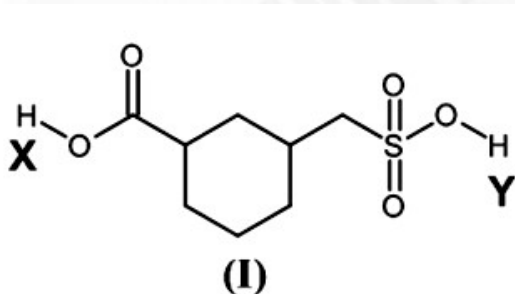


53. IUPAC name of the following compound is



- A) 5-(but-3-enyl)nona-3,6-diene-1-yne-5-ol
 B) 5-(but-1-enyl)nona-3,6-diene-1-yne-5-ol
 C) 5-(but-2-en-3-ynyl)nona-1,6-dien-5-ol
 D) 5-(but-3-enyl)nona-3,6-dien-1-yn-5-ol

54. Ionizable protons in the following compounds are marked as X and Y.



More acid proton is

- A) I-X; II-X; III-X ;IV-X
 B) I-Y; II-X; III-X ;IV-Y
 C) I-Y; II-X; III-Y ;IV-X
 D) I-X; II-Y; III-Y ;IV-X





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